Project Milestone 2

May 24, 2026

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DSC 680

Business Problem

Retaining talent is an important aspect of a business to ensure that the business succeeds and does not lose talented employees to competitive businesses. Losing employees is also costly for companies in that they will need to search for and interview potential hires and spend time training and onboarding them, which all costs time and money. This money can be saved by ensuring that employees feel supported and enjoy their roles. Knowing if an employee is unhappy and may soon leave will give a company insight into what factors influence if they stay and help them determine which areas they need to increase their focus on to retain talent. For this project, I will analyze a human resources dataset to predict whether an employee will be voluntarily terminated from a company.

Background/History

There are a lot of benefits to HR analytics that are more widely available today than ever. The benefits include improvements to hiring systems, making better training programs, creating a more effective payroll system and improving retention strategies (Visier, n.d.). In the past it was difficult to tell which strategies were most effective at keeping employees and there was not much data available to support any claims or an easy way to keep track of what may have caused employees to leave. With the increase in the amount of workforce metric data collected, it is easier to look at various metrics that were not even previously considered in the past that may predict if an employee will voluntarily terminate their employment.

Data Explanation

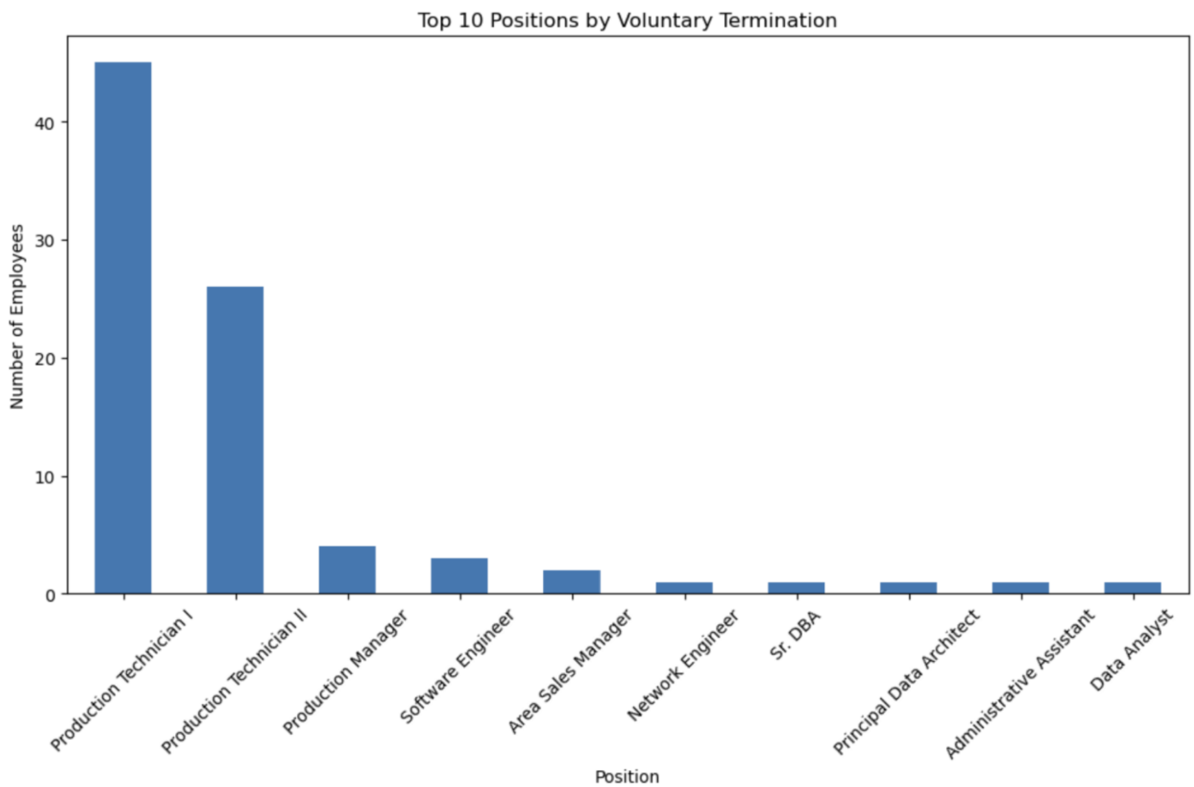
The data that I am using for this project comes from a Kaggle dataset that was created for teaching data visualization and regression techniques to HR students. This data was created to mimic data that would be found at a company which is generally hard to come by as publicly available for usage. Although this data is synthetic, it provides a good representation of data that would be found in HR including information on employee gender, marital status, salary, position, date of birth and more. The data set also contains information based on employee reviews, and whether the employee terminated voluntarily or not and the reason for termination.

Methods

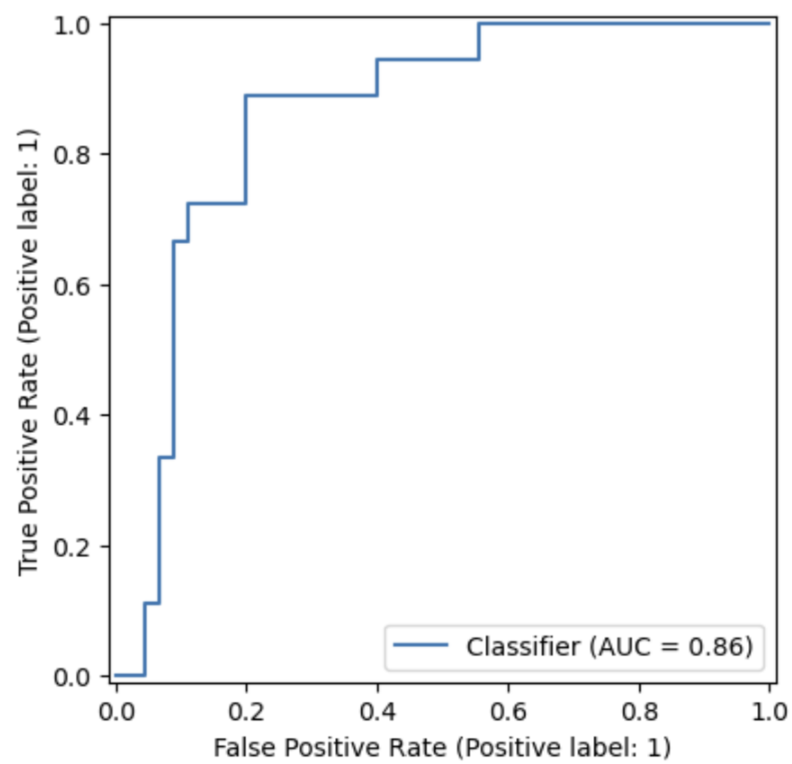
For this project, I will clean the data sets and will use exploratory data analysis to learn more about the data and see if there are any outliers or potential correlations between features. A great way to explore the data is with visualizations, and I will plan to use Python to create images that will give more insight to the data. Predictive modeling will be used to determine the likelihood of a person voluntarily terminating their employment with the company using logistic regression since it can be easily interpreted by HR professionals. Model performance will be analyzed using a ROC curve, precision/recall and F1 scores. Analysis of feature importance will be used to determine which measures are most influential and predictive of employees voluntarily leaving their positions.

Analysis

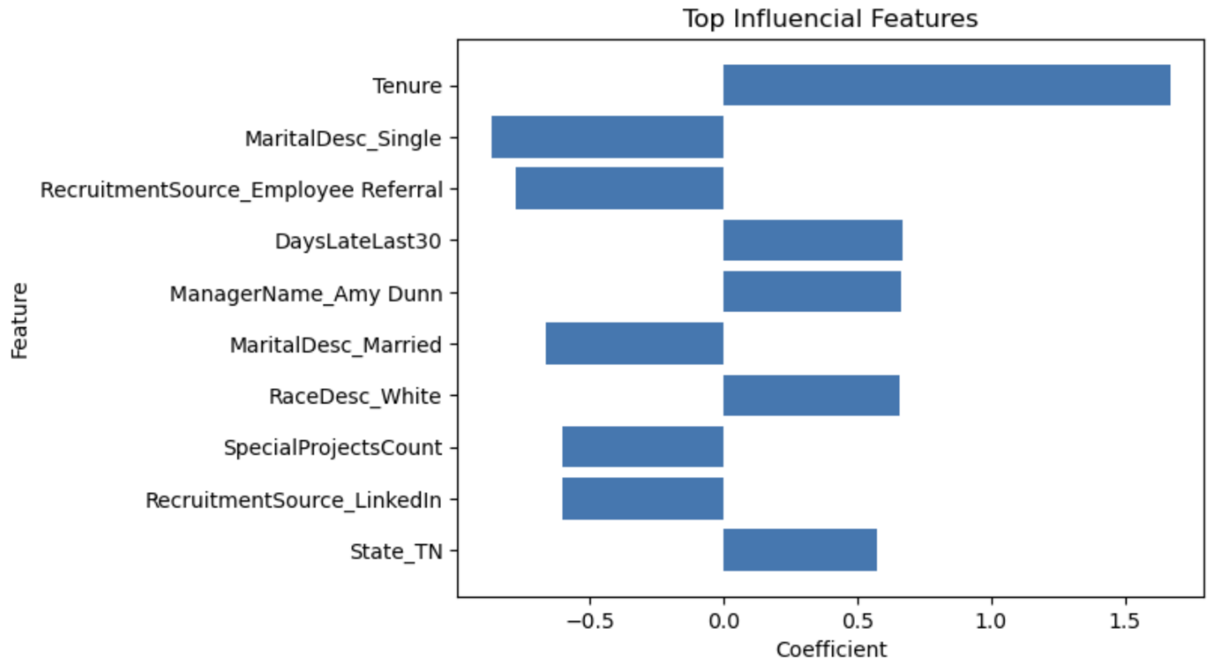
Before creating the model, I analyzed the data to see which employee positions had the most people who voluntarily left within this dataset as shown in the chart below. The production technical roles had the most voluntary terminations by far, so I would expect to see that role be an influence to the model.



The model performed fairly well considering the small dataset that was used and had a recall of 0.78. This means that the model was able to identify 78% of employees that voluntarily left. The accuracy came out to be 76% and the precision is 88%. The precision is a bit low and predicted more people that may leave that did not, although they still might in the future, but the recall is more important in this scenario. The ROC curve below is choppy due to the small dataset however and AUC of 0.82 shows that the model was fairly effective at finding which employees are at risk of leaving without having too many false positives.



The features that had the most influence on determining if an employee is likely to stay or leave are show in the graphic below. According to this chart, tenure had a big influence on voluntary termination along with an increase in number of late days and having a manager Amy Dunn. Employees who were hired from employee referrals were more likely to still be employed.



Conclusion

This model performed fairly well considering the small dataset however I would continue to add more features in the future to try and increase the model accuracy and recall. This model showed how influential the specific manager is on whether employees leave the company or are more likely to stay along with which positions have higher turnover.

Assumptions

An assumption that I made during this project is that the synthetic HR data accurately reflects data that would be found in an organization.

Limitations

One of the limitations to this project is the small sample size in that there are only 311 rows of data. More data would allow for a better analysis and a small data set might be more affected by noise than a larger data set would.

Challenges

One challenge I experienced is a class imbalance in the data in that there are only a few people that have voluntarily terminated their employment compared to the number of people who have stayed. This will have to be addressed by adding class weights.

Future Uses/Additional Applications

This model can be used in other situations where it is desirable to know if someone is likely to leave an organization or not even unrelated to HR. This could even include athletes leaving a sports team to go to another one or retiring.

Recommendations

I would recommend collecting more data since there are a limited number of rows in the data used to train this model and would also see if there are any other features that can be added to the dataset. For instance, if an organization collects any additional information about their employees that can be added to the model to see how it may affect employee termination. I would also recommend making changes based on the results of the most important features and seeing if more training could be provided for some managers who had employees leave more often or to see if they can make a position better that many employees leave as well.

Implementation Plan

To implement the model, data should be collected that is similar to the data used in this model and even could include new information. The data should be cleaned and the model should be tested for a while before being fully implemented to see how well it predicts future terminations at a company. Once the model is implemented, the data should be updated frequently and the results should provide insight into what changes may need to be made at the company in order to retain their employees.

Ethical Assessment

Some ethical considerations include that personally identifiable information should not be included in the analysis, including people's names, dates of birth or employee IDs in order to protect their privacy. The information discovered in this model should also not be used to discipline employees, but rather to see how the organization can improve to help keep talented employees. This information discovered is also only specific to this data set and the results may change based on the organization.

10 Questions an Audience May Ask

1. Why did you choose logistic regression for this model?
2. How did you handle a class imbalance?
3. What features were the most influential on the model?
4. How can this model help companies?
5. Is recall or precision more important in this problem?
6. What improvements would you make to the model?
7. What are the ethical concerns related to this project?
8. How does the size of the dataset affect the results?
9. How did you determine how to best evaluate the model?
10. Should this model be used for promotion or termination?

References

Huebner, R. (2023). *Human resources data set* [Data set]. Kaggle. https://www.kaggle.com/datasets/rhuebner/human-resources-data-set

Visier. (n.d.). *HR analytics*. Visier. https://www.visier.com/hr-analytics/

APPENDIX

Rhianna Ruggiero

Humamn Resources Data Model

In [662…

```
import pandas as pd
import numpy as np

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import (
    classification_report,
    confusion_matrix,
    roc_auc_score,
    RocCurveDisplay
)

import matplotlib.pyplot as plt
```

In [664…

```
#read data into python
df = pd.read_csv("HRDataset_v14.csv")
```

In [666…

```
df.head()
```

Out[666…

	Employee_Name	EmpID	MarriedID	MaritalStatusID	GenderID	EmpStatusID	DeptID
0	Adinolfi, Wilson K	10026	0	0	1	1	5
1	Ait Sidi, Karthikeyan	10084	1	1	1	5	3
2	Akinkuolie, Sarah	10196	1	1	0	5	5
3	Alagbe,Trina	10088	1	1	0	1	5
4	Anderson, Carol	10069	0	2	0	5	5

5 rows × 36 columns

In [668…

```
#check for missing values
df.isnull().sum()
```

Out[668…

```
Employee_Name                  0
EmpID                          0
MarriedID                      0
MaritalStatusID                0
GenderID                       0
EmpStatusID                    0
DeptID                         0
PerfScoreID                    0
FromDiversityJobFairID         0
Salary                         0
Termd                          0
PositionID                     0
Position                       0
State                          0
Zip                            0
DOB                            0
Sex                            0
MaritalDesc                    0
CitizenDesc                    0
HispanicLatino                 0
RaceDesc                       0
DateofHire                     0
DateofTermination            207
TermReason                     0
EmploymentStatus               0
Department                     0
ManagerName                    0
ManagerID                      8
RecruitmentSource              0
PerformanceScore               0
EngagementSurvey               0
EmpSatisfaction                0
SpecialProjectsCount           0
LastPerformanceReview_Date     0
DaysLateLast30                 0
Absences                       0
dtype: int64
```

It makes sense that there would be missing values for date of termination since those employees are likely still with the company, however this column will also cause data leakage and will be removed. Manager ID can also be dropped since the same information is essentially contained in Manager Name.

In [671…

```
#create a column that states whether an employee voluntarily terminated or not (sin
df["VoluntarilyLeft"] = np.where(
    df["EmploymentStatus"].str.contains("Voluntarily", case=False, na=False),
    1,
    0
)
```

In [673…

```
df.head()
```

Out[673...

	Employee_Name	EmpID	MarriedID	MaritalStatusID	GenderID	EmpStatusID	DeptID
0	Adinolfi, Wilson K	10026	0	0	1	1	5
1	Ait Sidi, Karthikeyan	10084	1	1	1	5	3
2	Akinkuolie, Sarah	10196	1	1	0	5	5
3	Alagbe,Trina	10088	1	1	0	1	5
4	Anderson, Carol	10069	0	2	0	5	5

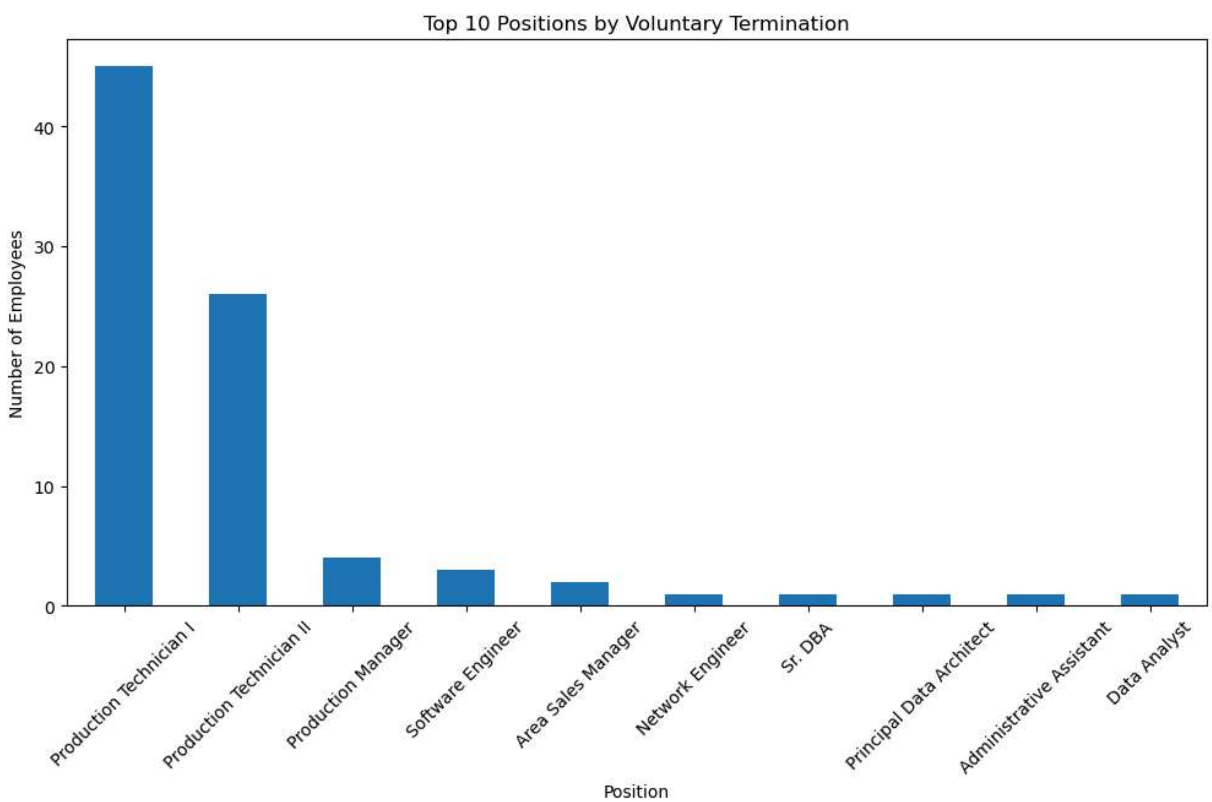
5 rows × 37 columns

In [675...

```
df["VoluntarilyLeft"] = df["EmploymentStatus"].str.contains("Voluntarily").astype(i

df.groupby("Position")["VoluntarilyLeft"].sum().sort_values(ascending=False).head(1

plt.title("Top 10 Positions by Voluntary Termination")
plt.ylabel("Number of Employees")
plt.xticks(rotation=45)
plt.show()
```



In [677...

```
#drop unnecessary columns that either cause leakage or don't provide useful informa
drop_cols = [
    "Employee_Name",
    "EmpID",
    "TermReason",
```

```
    "DateofTermination",
    "EmploymentStatus",
    "LastPerformanceReview_Date",
    "ManagerID",
    "PositionID",
    "DeptID",
    "EmpStatusID",
    "Zip",
    "DOB",
    "MaritalStatusID",
    "MarriedID",
    "Termd"
    #last performance review date could indicate that an employee left since they w
]

df = df.drop(columns=drop_cols)
```

In [679…

```
print(df.dtypes)
```

```
GenderID                     int64
PerfScoreID                  int64
FromDiversityJobFairID       int64
Salary                       int64
Position                    object
State                       object
Sex                         object
MaritalDesc                 object
CitizenDesc                 object
HispanicLatino              object
RaceDesc                    object
DateofHire                  object
Department                  object
ManagerName                 object
RecruitmentSource           object
PerformanceScore            object
EngagementSurvey           float64
EmpSatisfaction              int64
SpecialProjectsCount         int64
DaysLateLast30               int64
Absences                     int64
VoluntarilyLeft              int32
dtype: object
```

In [681…

```
df["DateofHire"].dtype
```

Out[681…

```
dtype('O')
```

Date of hire could be an important column to see if length of time in the role has an effect, so I will change the date to a tenure column.

In [684…

```
df["DateofHire"] = pd.to_datetime(df["DateofHire"])

df["Tenure"] = (pd.Timestamp.today() - df["DateofHire"]).dt.days

df = df.drop(columns=["DateofHire"])
```

In [686...

```
df.head()
```

Out[686...

	GenderID	PerfScoreID	FromDiversityJobFairID	Salary	Position	State	Sex	MaritalI
0	1	4	0	62506	Production Technician I	MA	M	Si
1	1	3	0	104437	Sr. DBA	MA	M	Mar
2	0	3	0	64955	Production Technician II	MA	F	Mar
3	0	3	0	64991	Production Technician I	MA	F	Mar
4	0	3	0	50825	Production Technician I	MA	F	Divo

5 rows × 22 columns

In [688...

```
#many columns are categorical and need to be encoded
df = pd.get_dummies(df, drop_first=True)
```

In [690...

```
df = df.astype(int, errors='ignore')
```

In [692...

```
X = df.drop("VoluntarilyLeft", axis=1)
y = df["VoluntarilyLeft"]
```

In [694...

```
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42,
    stratify=y
    #stratify used for class imbalance
)
```

In [696...

```
from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
```

In [698...

```
model = LogisticRegression(
    class_weight='balanced',
    solver='lbfgs',
    max_iter=1000
)
```

```python
model.fit(X_train_scaled, y_train)
```

Out[698…

```
LogisticRegression
LogisticRegression(class_weight='balanced', max_iter=1000)
```

In [700…

```python
y_pred = model.predict(X_test_scaled)
y_prob = model.predict_proba(X_test_scaled)[:, 1]
```

In [702…

```python
print(classification_report(y_test, y_pred))
```

```
              precision    recall  f1-score   support

           0       0.88      0.78      0.82        45
           1       0.57      0.72      0.63        18

    accuracy                           0.76        63
   macro avg       0.72      0.75      0.73        63
weighted avg       0.79      0.76      0.77        63
```

In [704…

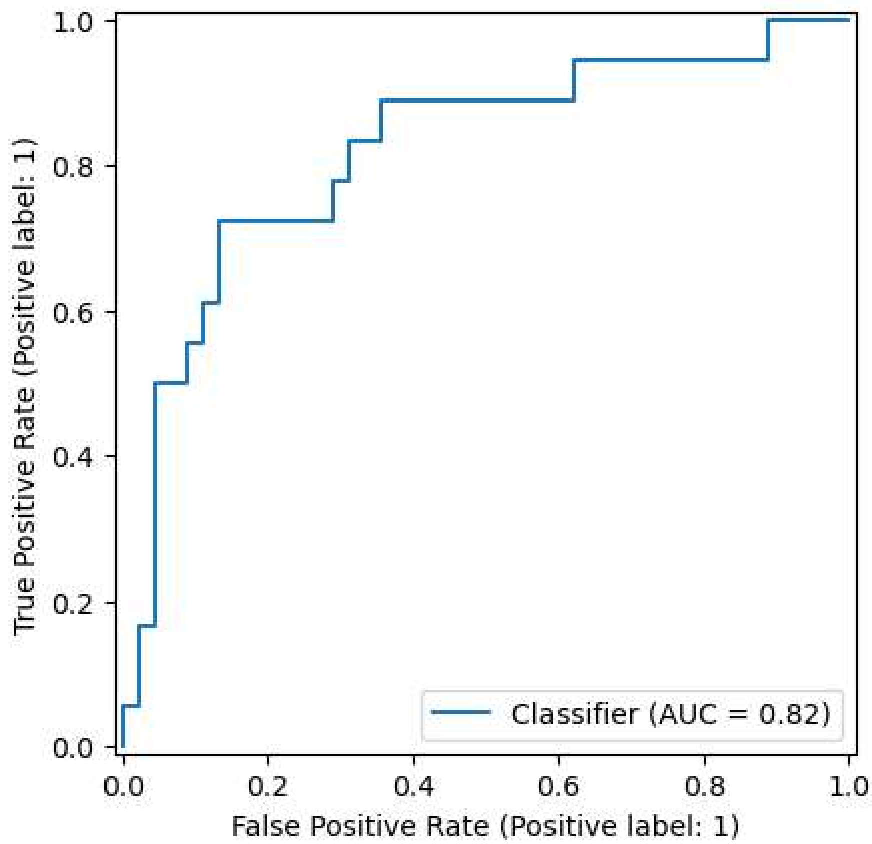
```python
print(confusion_matrix(y_test, y_pred))
```

```
[[35 10]
 [ 5 13]]
```

In [706…

```python
RocCurveDisplay.from_predictions(y_test, y_prob)

plt.show()
```



In [708…

```
coefficients = pd.DataFrame({
    "Feature": X.columns,
    "Coefficient": model.coef_[0]
})

print(coefficients.sort_values(by="Coefficient", ascending=False))
```

```
                               Feature  Coefficient
9                               Tenure     1.667250
7                         DaysLateLast30     0.669456
88                    ManagerName_Amy Dunn     0.664280
82                        RaceDesc_White     0.658006
62                              State_TN     0.570966
..                                 ...          ...
112            RecruitmentSource_LinkedIn    -0.598926
6                   SpecialProjectsCount    -0.600584
69                   MaritalDesc_Married    -0.660127
109  RecruitmentSource_Employee Referral    -0.772638
71                    MaritalDesc_Single    -0.865414

[119 rows x 2 columns]
```

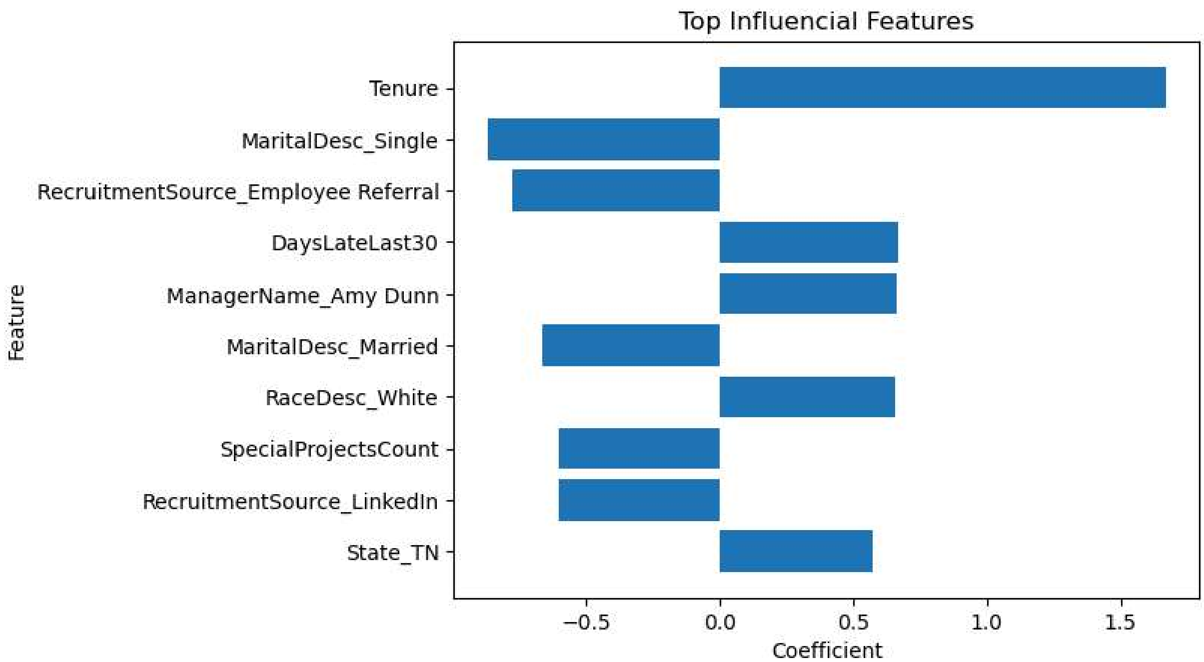
In [710…

```
# top positive and negative drivers
top_coeffs = coefficients.sort_values(
    by="Coefficient",
    key=abs,
    ascending=False
).head(10)

plt.barh(top_coeffs["Feature"], top_coeffs["Coefficient"])
```

```
plt.xlabel("Coefficient")
plt.ylabel("Feature")
plt.title("Top Influencial Features")
plt.gca().invert_yaxis()

plt.show()
```




```
In [ ]:
```